

Mean Shift through Hilbert's Projective Metric

A Sinkhorn-Style Convergence Argument

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July 2026

Abstract

Blurring mean shift repeatedly replaces every particle by a kernel-weighted average of the current cloud. For a strictly positive kernel, each update is therefore multiplication by a positive row-stochastic matrix, although this matrix changes with the particle positions. This note gives a short convergence proof based on the same projective contraction mechanism used for Sinkhorn scaling. Birkhoff's theorem contracts Hilbert's metric on the positive cone; linearizing this metric at the constant ray gives Hopf's oscillation seminorm, whose exact Markov contraction factor is Dobrushin's coefficient. A uniform projective bound consequently yields geometric consensus for discrete mean shift and exponential consensus for its continuous-time limit. The argument also makes precise what the analogy with Sinkhorn does, and does not, imply.

1 Blurring mean shift as state-dependent Markov averaging

Classical mean shift moves a query point toward a kernel-weighted barycenter of a *fixed* data set [5, 3]. Here we study the blurring, or self-consistent, variant: every particle moves and the averaging measure is recomputed after each update [2]. This distinction is essential because only the self-consistent dynamics is a consensus system.

Let $a_i > 0$, with $\sum_{i=1}^n a_i = 1$, and let $X = (x_1, \dots, x_n)^\top \in (\mathbb{R}^d)^n$. Given a positive kernel $K : \mathbb{R}^d \times \mathbb{R}^d \rightarrow (0, +\infty)$, define

$$(K_X)_{ij} = K(x_i, x_j), \quad A = \text{diag}(a_1, \dots, a_n), \quad P_X = \text{diag}(K_X A)^{-1} K_X A. \quad (1)$$

The entries of P_X are

$$(P_X)_{ij} = \frac{a_j K(x_i, x_j)}{\sum_{\ell=1}^n a_\ell K(x_i, x_\ell)}. \quad (2)$$

Thus $P_X > 0$ and $P_X \mathbf{1} = \mathbf{1}$. Its i th row is the probability law used to average the cloud from the viewpoint of x_i .

For a damping parameter $0 < \tau \leq 1$, discrete blurring mean shift is

$$X^{k+1} = (1 - \tau)X^k + \tau P_{X^k} X^k. \quad (3)$$

The choice $\tau = 1$ is the full barycentric update. Its continuous-time limit is the nonautonomous consensus system

$$\dot{X}(t) = (P_{X(t)} - I)X(t). \quad (4)$$

Every new position and every instantaneous target $P_X X$ belongs to the convex hull of the current positions. Consequently the convex hull is nested along either dynamics. In particular, if

$$C_0 = \text{conv}\{x_1^0, \dots, x_n^0\},$$

then all subsequent configurations remain in the compact convex set C_0 .

2 From Hilbert contraction to oscillation contraction

The projective argument is most transparent for a fixed positive matrix. We first recall the metric used in the convergence analysis of Sinkhorn scaling.

Definition 2.1 (Hilbert metric and oscillation). For $u, v \in \mathbb{R}_{++}^n$, Hilbert's projective metric is

$$d_H(u, v) = \max_i \log \frac{u_i}{v_i} - \min_i \log \frac{u_i}{v_i}. \quad (5)$$

For $z \in \mathbb{R}^n$, Hopf's oscillation seminorm is

$$\text{osc}(z) = \max_i z_i - \min_i z_i. \quad (6)$$

Both quantities ignore the constant direction: d_H is invariant under positive rescaling, while $\text{osc}(z + c\mathbf{1}) = \text{osc}(z)$.

The two geometries are related exactly by logarithmic coordinates,

$$d_H(e^z, e^{z'}) = \text{osc}(z - z'), \quad (7)$$

and infinitesimally by

$$d_H(\mathbf{1} + sz, \mathbf{1}) = |s| \text{osc}(z) + o(|s|) \quad (s \rightarrow 0). \quad (8)$$

For a positive matrix L , introduce its projective cross-ratio and Birkhoff factor

$$\eta(L) = \max_{i,\ell,j,r} \frac{L_{ij}L_{\ell r}}{L_{ir}L_{\ell j}}, \quad q(L) = \frac{\sqrt{\eta(L)} - 1}{\sqrt{\eta(L)} + 1} = \tanh\left(\frac{\log \eta(L)}{4}\right). \quad (9)$$

Theorem 2.2 (Birkhoff contraction). *For every positive matrix L and all $u, v > 0$,*

$$d_H(Lu, Lv) \leq q(L)d_H(u, v). \quad (10)$$

This is Birkhoff's cone-contraction theorem [1]. Franklin and Lorenz used precisely this result to obtain geometric convergence of positive matrix scaling [4].

For mean shift, the state coordinates are not positive vectors and the matrix changes at every iteration. The correct bridge is to linearize Theorem 2.2 at the fixed projective ray generated by $\mathbf{1}$.

Proposition 2.3 (Hilbert contraction linearizes to consensus contraction). *Let $P > 0$ be row-stochastic. Then, for every $z \in \mathbb{R}^n$,*

$$\text{osc}(Pz) \leq q(P) \text{osc}(z). \quad (11)$$

Proof. Apply Theorem 2.2 to $u_s = e^{sz}$ and $v = \mathbf{1}$. Since $P\mathbf{1} = \mathbf{1}$,

$$d_H(Pe^{sz}, \mathbf{1}) \leq q(P)d_H(e^{sz}, \mathbf{1}) = q(P)|s| \text{osc}(z).$$

Moreover, $Pe^{sz} = \mathbf{1} + sPz + O(s^2)$, so $\log(Pe^{sz}) = sPz + O(s^2)$. Divide by $|s|$ and let $s \rightarrow 0$ to obtain (11). $\square$

The exact linear contraction factor has a classical probabilistic expression.

Proposition 2.4 (Dobrushin coefficient). *For a row-stochastic matrix P , define*

$$\delta(P) = \frac{1}{2} \max_{i,\ell} \sum_j |P_{ij} - P_{\ell j}| = 1 - \min_{i,\ell} \sum_j \min(P_{ij}, P_{\ell j}). \quad (12)$$

Then

$$\text{osc}(Pz) \leq \delta(P) \text{osc}(z) \quad \text{for every } z \in \mathbb{R}^n, \quad (13)$$

and $\delta(P)$ is the smallest possible constant. If $P > 0$, then

$$\delta(P) \leq q(P) < 1. \quad (14)$$

Proof. For two rows p and p' , remove their common mass $r_j = \min(p_j, p'_j)$. The two residual positive measures have the same total mass

$$1 - \sum_j r_j = \frac{1}{2} \sum_j |p_j - p'_j|.$$

Their integrals against z can therefore differ by at most this mass times $\text{osc}(z)$. Taking the largest difference between two output coordinates gives (13); choosing z as an indicator of the positive part of a maximizing row difference proves sharpness. Inequality (14) follows from Proposition 2.3. This relation between Dobrushin contraction, Hopf oscillation and the infinitesimal Hilbert metric is developed systematically in [6]. $\square$

3 Projective convergence of blurring mean shift

We now apply the preceding fixed-matrix result to the state-dependent matrices P_X . The key observation is that row normalization and the particle weights do not alter projective cross-ratios. Indeed,

$$\frac{(P_X)_{ij}(P_X)_{\ell r}}{(P_X)_{ir}(P_X)_{\ell j}} = \frac{K(x_i, x_j)K(x_\ell, x_r)}{K(x_i, x_r)K(x_\ell, x_j)}. \quad (15)$$

This is the projective counterpart of the fact that multiplication by positive diagonal matrices is an isometry of Hilbert's metric.

For a configuration X , write

$$D(X) = \max_{i,\ell} \|x_i - x_\ell\|. \quad (16)$$

For every unit vector $\theta \in \mathbb{R}^d$, let $z_\theta = X\theta \in \mathbb{R}^n$. The elementary support-function identity

$$D(X) = \sup_{\|\theta\|=1} \text{osc}(z_\theta) \quad (17)$$

reduces contraction in $\mathbb{R}^d$ to scalar oscillation contraction.

Theorem 3.1 (Sinkhorn-style consensus bound). *Assume that K is Lipschitz and strictly positive on $C_0 \times C_0$, and set*

$$m = \min_{C_0 \times C_0} K > 0, \quad M = \max_{C_0 \times C_0} K < +\infty, \quad q = \frac{M - m}{M + m} < 1. \quad (18)$$

Then the following statements hold.

1. For the damped discrete dynamics (3), let

$$q_\tau = 1 - \tau(1 - q) < 1. \quad (19)$$

Then

$$D(X^k) \leq q_\tau^k D(X^0). \quad (20)$$

There exists $x_\infty \in C_0$ such that every particle converges to x_∞ , with

$$\max_i \|x_i^k - x_\infty\| \leq \frac{\tau D(X^0)}{1 - q_\tau} q_\tau^k = \frac{D(X^0)}{1 - q} q_\tau^k. \quad (21)$$

2. For the continuous dynamics (4),

$$D(X(t)) \leq e^{-(1-q)t} D(X(0)). \quad (22)$$

There exists $x_\infty \in C_0$ such that

$$\max_i \|x_i(t) - x_\infty\| \leq \frac{D(X(0))}{1 - q} e^{-(1-q)t}. \quad (23)$$

Proof. The finite-dimensional vector field in (4) is Lipschitz on C_0^n , and convex-hull invariance keeps every configuration inside C_0 ; hence the continuous dynamics is globally well posed. From (15) and (18),

$$\eta(P_X) \leq \left(\frac{M}{m}\right)^2, \quad q(P_X) \leq \frac{M - m}{M + m} = q$$

uniformly along the dynamics.

For the discrete update and each direction θ , Proposition 2.3 gives

$$\begin{aligned} \text{osc}(z_\theta^{k+1}) &\leq (1 - \tau) \text{osc}(z_\theta^k) + \tau \text{osc}(P_{X^k} z_\theta^k) \\ &\leq (1 - \tau + \tau q) \text{osc}(z_\theta^k) = q_\tau \text{osc}(z_\theta^k). \end{aligned}$$

Take the supremum over θ and iterate to prove (20). Since

$$\|x_i^{k+1} - x_i^k\| \leq \tau D(X^k),$$

every trajectory is Cauchy. The vanishing diameter forces all limits to coincide, and summing the geometric tail gives (21).

For continuous time, fix θ and write $z(t) = X(t)\theta$. Then

$$\dot{z}(t) = (P_{X(t)} - I)z(t).$$

The upper Dini derivative of the oscillation satisfies

$$D^+ \text{osc}(z(t)) \leq \text{osc}(P_{X(t)} z(t)) - \text{osc}(z(t)) \leq -(1 - q) \text{osc}(z(t)).$$

Gronwall's lemma and (17) give (22). Finally, $\|\dot{x}_i(t)\| \leq D(X(t))$; integrating this bound proves convergence and (23). $\square$

The theorem uses the easily read bound $m \leq K \leq M$. The sharper intrinsic constant is

$$q_* = \sup_{X \in C_0^n} q(P_X), \quad (24)$$

which can replace q throughout whenever $q_* < 1$. The exact oscillation constant is smaller still: one may replace $q(P_X)$ by $\delta(P_X)$ at each step. The Hilbert bound is useful because its cross-ratio formula is explicit and invariant under all row and column scalings.

4 Gaussian mean shift and an adaptive rate

For the Gaussian kernel

$$K_\varepsilon(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\varepsilon}\right), \quad (25)$$

the current convex hull has diameter D , so $M = 1$ and $m = e^{-D^2/(2\varepsilon)}$. The corresponding projective factor is

$$q(D) = \frac{1 - e^{-D^2/(2\varepsilon)}}{1 + e^{-D^2/(2\varepsilon)}} = \tanh\left(\frac{D^2}{4\varepsilon}\right). \quad (26)$$

Corollary 4.1 (Adaptive Gaussian contraction). *For the full discrete update $\tau = 1$,*

$$D(X^{k+1}) \leq \tanh\left(\frac{D(X^k)^2}{4\varepsilon}\right) D(X^k) \leq \frac{D(X^k)^3}{4\varepsilon}. \quad (27)$$

For the continuous-time dynamics,

$$D^+ D(X(t)) \leq - \left[1 - \tanh\left(\frac{D(X(t))^2}{4\varepsilon}\right) \right] D(X(t)). \quad (28)$$

Thus strict positivity gives a global geometric rate, while contraction becomes much stronger once the cloud is narrow relative to the bandwidth. In particular, the projective estimate predicts a cubic local bound for the full discrete Gaussian update. For a narrow kernel and a widely separated initial cloud, however, $q(D_0)$ is extremely close to one; a long-lived multimodal transient is therefore compatible with eventual one-point consensus.

5 Measure-valued formulation

The same mechanism does not depend on a finite particle representation. For $\alpha \in \mathcal{P}(C_0)$, define the Markov operator

$$(P_\alpha f)(x) = \frac{\int_{C_0} K(x, y) f(y) d\alpha(y)}{\int_{C_0} K(x, y) d\alpha(y)}. \quad (29)$$

It fixes constants, and the integral-operator version of Birkhoff's theorem gives

$$\text{osc}_{C_0}(P_\alpha f) \leq q \text{osc}_{C_0}(f), \quad q = \frac{M - m}{M + m}. \quad (30)$$

The associated mean-shift field is

$$V[\alpha](x) = P_\alpha \text{Id}(x) - x, \quad (31)$$

and its characteristic evolution solves

$$\partial_t \alpha_t + \nabla \cdot (\alpha_t V[\alpha_t]) = 0. \quad (32)$$

If K is Lipschitz and positive on C_0^2 , the denominator in (29) is uniformly positive, so the characteristic flow is well posed and its convex hull remains in C_0 . Applying (30) to $f_\theta(x) = \langle \theta, x \rangle$ gives exactly the directional-width argument in Theorem 3.1. Consequently

$$\text{diam}(\text{supp } \alpha_t) \leq e^{-(1-q)t} \text{diam}(\text{supp } \alpha_0), \quad (33)$$

and the characteristic trajectories converge to a common point. Hence α_t converges to a Dirac mass, with the same exponential estimate in $\mathcal{W}_\infty$.

6 What is shared with Sinkhorn, and what is not

For a fixed positive matrix K , one Sinkhorn cycle alternates

$$u^+ = \frac{a}{Kv}, \quad v^+ = \frac{b}{K^\top u^+}. \quad (34)$$

Multiplication by a positive diagonal vector and entrywise inversion are isometries of d_H . Each kernel multiplication contracts by $q(K)$, so a full Sinkhorn cycle contracts projective scaling vectors by $q(K)^2$ [4].

Mean shift uses the same ingredients in a different order. It first forms the positive kernel matrix K_X , then row-normalizes it into the Markov matrix P_X . Diagonal normalization preserves the projective cross-ratios, while $P_X \mathbf{1} = \mathbf{1}$ makes the constant ray a fixed point. Linearizing Hilbert contraction at this ray yields contraction of spatial oscillations. The common mechanism can therefore be summarized as follows:

	Sinkhorn	Blurring mean shift
positive object	fixed Gibbs kernel K	state-dependent kernel K_X
normalization	two prescribed marginals	row sums equal to one
contracted quantity	scaling rays in d_H	coordinate widths in osc
rate mechanism	Birkhoff, twice per cycle	infinitesimal Birkhoff/Dobrushin
limit	balanced coupling	consensus Dirac mass

Three qualifications prevent overinterpreting the analogy.

1. The estimate contracts the diameter *within one evolving cloud*. Because P_X depends on X , it does not by itself show that the nonlinear map $X \mapsto P_X X$ contracts the distance between two different configurations.
2. Strict positivity is decisive. If K can vanish, then the projective diameter can be infinite and the Birkhoff factor can be one. Bounded-confidence models can then converge to several disconnected clusters rather than one consensus point [7].
3. The result concerns blurring mean shift. In classical nonblurring mean shift the reference data distribution is frozen; different initial queries may converge to different modes even when the kernel is everywhere positive.

The useful conceptual conclusion is therefore precise: positive-kernel mean shift and Sinkhorn are not the same fixed-point iteration, but their linear rates are governed by the same projective loss-of-memory mechanism. Sinkhorn uses Hilbert contraction globally on scaling rays; mean shift uses its infinitesimal Hopf–Dobrushin form on spatial coordinates.

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